Self State Aware Autonomy for Disconnected Driving

Onboard decisions during network loss

Wonsik Choi and Jeongin Choi | Independent Researchers

janefather@gmail.com | cjeongin024@gmail.com

Wonsik Choi ORCID 0009-0001-4263-9772 | Proposal v0.1 | 13 October 2026

Technical review proposal for Tesla autonomous driving and robotics engineering teams

Assessment stage	Scope of this proposal
Inputs and validation basis	Availability of onboard sensors, models and controllers; connectivity, resources, roads and safety constraints.
Architecture transformation	Update self model → generate onboard plan → authorize execution → drive or respond safely → retain residue.
Outputs	A destination route or route to a candidate refuge selected from current conditions during network loss.
Falsification conditions	Treat unverified external information as certain; authorize invalid routes; violate budgets; interrupt essential control.

When a network connection fails during autonomous driving, a vehicle can still evaluate current conditions if its onboard sensors, decision models and controllers remain available. This proposal addresses an execution architecture that maintains its own observation, resource and action capabilities alongside connectivity status, and recomposes plans when road conditions change.

Operational definition of elementary consciousness

In this proposal, elementary consciousness means maintaining a self model of owned equipment and boundaries, current resources, accumulated damage, decision evidence, and available actions, and changing plans in response to that state. Whether subjective experience exists is outside the scope of this execution experiment.

Humans specify available equipment and the physical constraints of basic actions. The planner composes actions from current observations and its self model. Evaluation distinguishes this capability from writing a complete response sequence for each individual fault combination.

Core contribution

Record network loss as the loss of an external service. Do not automatically infer that onboard perception is also unavailable. Continue when available onboard information supports the driving constraints, and compose a feasible safety response when it does not. The particular implementation must declare its connectivity dependencies and permitted operating conditions.

Vehicle self model and execution authorization

Element	Execution specification
Current inputs	Attach provenance, timestamps and validity periods to observations and equipment health.
Self model	Distinguish owned equipment, external services, resources, damage and unresolved conditions.
Action generation	Generate candidate plans from currently available basic actions and the goal.
Shared execution path	Apply the same proposal, inspection, authorization and commit contract to every candidate.
Resource gate	Check energy, peak power, processing latency and record capacity separately.
Residue and ledger	Carry faults, pending tasks, remaining resources and decision evidence into the next decision.

The energy example uses $E_{after} = E_{before} - E_{task} - E_{overhead}$. Discretionary work requires $E_{after} \geq E_{reserve}$. Resources with different units are not added without a declared conversion. Planning, monitoring, recording and essential control costs belong in the resource ledger.

Recheck the input revision and equipment state immediately before authorization. A deferred proposal does not overwrite the execution state, although the environment and resource consumption continue to evolve. Revalidate pending commands after network or sensor recovery. Design the underlying controller separately to maintain mission specific survival functions.

Distinguishing three operating conditions

First, if connectivity alone is lost and onboard observations remain valid, compute routes onboard. Second, if valid onboard observations reveal a new obstacle, invalidate the affected route and replan.

Third, if visibility, sensing or control capabilities fail to meet required safety constraints, select the mission specific minimal risk response. Do not collapse these conditions into a single network status.

Roles of driving control and planning

A physical application must connect continuous collision avoidance, lanes and speed, braking distance, road friction, other vehicles and observation errors. The existence of a destination route alone does not authorize safe driving. Independently validate the movement to a refuge; do not assume a safe stopping location is always reachable.

Tesla public technical materials describe perception, planning and control for vehicles and robots, efficient inference hardware and evaluation infrastructure. This proposal supplies execution contracts that could connect to such capabilities; it does not claim knowledge of Tesla internal architecture or network loss policies. [2]

Executed route change during network loss

In a 5×5 grid, the vehicle starts at (0,2) and its destination is (4,2). The initial shortest route has four steps. After connectivity is lost, an onboard sensor observes that (2,2) is blocked. Perception remains valid within the grid model, and the vehicle computes a different route onboard.

The A* planner generates a six step detour from the current map. Movement costs 1 Wh per step, route processing costs 0.5 Wh, and the reserve is 4 Wh. Candidate refuges (1,1) and (3,1) are map inputs; their routes are computed from current conditions. These values and the grid are not actual road or vehicle specifications.

Current energy	Plan and authorization	Computed result
20 Wh	Authorize the six step detour	13.5 Wh after movement
10 Wh	Deny the six step destination route	Would leave 3.5 Wh and violate reserve
10 Wh	Authorize the two step refuge route	7.5 Wh after movement

Including 0.2 Wh for one essential control interval gives final balances of 13.3 Wh and 7.3 Wh, respectively. The invalidated initial route is denied in both cases. Even under the same disconnected conditions, current resources change which actions are authorized.

What execution established

The included demo.py generates destination and refuge routes with A* and records resource and input revision checks. Its self state output preserves onboard equipment availability, unavailable external service, remaining energy, the selected goal and unresolved items. No list of completed scenario specific routes is supplied.

This example implements high level route planning and resource accounting. It does not implement lanes, speed, braking, pedestrians or continuous collision avoidance. Reaching a candidate refuge in the grid is not reported as an actual vehicle safety or accident prevention result. Learned AI generalization requires a separate evaluation.

Comparative experiments and implementation stages

Evaluation condition	Metrics to assess together
Connectivity loss alone	Unnecessary mission interruption and continued onboard decisions
Loss and road changes	New route generation, invalid route denial and mission completion
Loss and sensor degradation	Recognition of perception limits, minimal risk response and erroneous authorization
Loss and resource shortage	Reserve compliance, essential control costs and feasible progress
Connection recovery	Revalidation of stale commands and consistency of states and records

Compare a prewritten response table, the same planner, and the planner with self model and execution contracts, using identical sensors, models, controllers and initial resources. Assess safety constraint violations, goal achievement, successful minimal risk responses, processing latency, energy and recording costs jointly. Overall denial rate alone does not establish superiority.

The executed reference case passed its invariants. Overall driving performance and computation savings remain uncomputed. The next stage connects these contracts to a vehicle simulator, injects connectivity loss, road changes and sensor faults, and evaluates combinations not used in development. Actual road deployment follows vehicle dynamics and hardware validation.

Proposal summary

We propose a vehicle execution architecture that maintains an explicit model of its own sensing, computing, control capability, available resources and unresolved conditions. During network loss, it uses onboard information to generate and authorize revised actions. The first evaluation compares continued driving and feasible minimal risk responses under held out combinations of connectivity loss, road changes, sensing degradation and resource limits.

Research provenance and technical references

[1] Wonsik Choi and Jeongin Choi. WRRA Core 2.0. 13 October 2026. Contracts for current state, common carrier, resource authorization, residue and provenance preservation are mapped to onboard vehicle execution. The original physical numbers and channel counts are not imposed on the vehicle.

[2] Tesla. AI and Robotics. https://www.tesla.com/AI
Reference checked on 13 October 2026. This public material provides an external comparison and basis for the application target. Tesla software is not integrated into the included code.